

Math Camp

Lecture 7

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Functions on $\mathbb{R}^n$

- We extend to a multivariate setting many of the notions and results obtained for univariate functions
- We are interested in functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$
- We exploit heavily the fact that $\mathbb{R}^n$ may be viewed as a *Hilbert space*
 - A complete normed vector space with a norm that may be derived from an inner product, $\|\mathbf{x}\|_2 \equiv \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$
- Unless otherwise stated the norm is the Euclidean norm, and we drop the subscript

Continuity

- The definition of continuity for functions from $\mathbb{R}^n$ to $\mathbb{R}^m$ — or between any two metric spaces — is exactly the same as in the univariate case
 - Continuous functions take convergent sequences to convergent sequences
- The results that continuity is preserved under point-wise addition and composition generalize to the multivariate case

Continuity

- The main result for univariate continuous functions was the intermediate value theorem
- Its most direct analog in the multivariate setting is that continuous functions respect *connectedness*
 - They take connected sets to connected sets
- In the univariate case connectedness is simple, a set is connected exactly when it is an interval

Fixed point theorems

- We state two *fixed point theorems*, which provide the kinds of existence results needed for the establishment of equilibria
- Recall that a compact subset of $\mathbb{R}^n$ is exactly a closed and bounded set
- $K \subset \mathbb{R}^n$ is *convex* provided that whenever $\mathbf{x}, \mathbf{y} \in K$ it follows that $\alpha\mathbf{x} + (1 - \alpha)\mathbf{y} \in K$ for all $\alpha \in (0, 1)$

Fixed point theorems

Brouwer's Theorem

Let $K \subset \mathbb{R}^n$ be compact and convex, and let $f : K \rightarrow K$ be continuous. Then there is a point $\mathbf{x} \in K$ such that $f(\mathbf{x}) = \mathbf{x}$.

- Quite powerful because its premise is weak and easy to check
- Like many existence theorems it suffers in the applicability of its conclusion
 - No mention of construction or of uniqueness of the fixed point

Contractions

- $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a *contraction* if there is a $\beta \in (0, 1)$ such that for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$

$$\|f(\mathbf{x}) - f(\mathbf{y})\| \leq \beta \|\mathbf{x} - \mathbf{y}\|$$

- ▶ Intuitively, f shrinks by at least a factor β the distance between points

Contraction mapping theorem

Contraction Mapping Theorem

Let $X \subset \mathbb{R}^n$ be closed. Suppose $f : X \rightarrow X$ is a contraction. Then there exists a unique $\mathbf{x}^* \in X$ such that $f(\mathbf{x}^*) = \mathbf{x}^*$. Furthermore, if $\mathbf{x}_0$ is any point in X then

$$f^n(\mathbf{x}_0) \equiv f(f^{n-1}(\mathbf{x}_0)) \rightarrow \mathbf{x}^*$$

- The theorem holds on any Banach space
- We get not only existence of a fixed point, but uniqueness
- We get a mechanism to approximate the solution, iterate on f
- This is the reason dynamic programming works in practice as well as in theory

Continuous functions on compact sets

Weierstrass Theorem

Let $K \subset \mathbb{R}^n$ be compact and $f : K \rightarrow \mathbb{R}$ be continuous. Then there exists $\mathbf{x}^* \in K$ such that

$$f(\mathbf{x}^*) = \sup_{\mathbf{x} \in K} f(\mathbf{x}) < \infty$$

- If f is a continuous real-valued function on a compact set then its supremum is finite and is attained by the function at some point in that set
 - Often called the *extreme value theorem*
 - It is the existence result standing behind every maximization problem we write down

Derivatives

- To generalize the derivative to functions from $\mathbb{R}^n$ to $\mathbb{R}^m$, two steps are needed
 - Define a *partial derivative*, from which ceteris paribus considerations arise
 - Define the derivative at a point as a *linear map*

Partial derivatives

- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The *partial derivative* of f with respect to x_i at $\mathbf{x} \in \mathbb{R}^n$, when it exists, is

$$f_i(\mathbf{x}) = \lim_{\Delta x_i \rightarrow 0} \frac{f(x_1, \dots, x_i + \Delta x_i, \dots, x_n) - f(\mathbf{x})}{\Delta x_i}$$

- ▶ The derivative of f with respect to x_i , holding all other coordinates constant
- ▶ The change in f that obtains when $\mathbf{x}$ changes only along the i th coordinate axis
- Often written $\partial f / \partial x_i$

Partial derivatives

- Higher-order partials are defined analogously
 - f_{ij} is the partial with respect to the j th component of the partial with respect to the i th component
 - Written $\partial^2 f / \partial x_i \partial x_j$
- If f_{ij} and f_{ji} are continuous then $f_{ij} = f_{ji}$
 - This is *Young's theorem*; the order of differentiation does not matter
 - A function of n variables has n^2 second-order partials, and Young's theorem makes the matrix of them symmetric

The derivative as a linear map

- Recall that a linear map from $\mathbb{R}$ to $\mathbb{R}$ is exactly and only multiplication
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Then $df = f'(x) dx$, so $f'(x)$ is the linear map that takes dx to df
- To make this formal let $T : \mathbb{R} \rightarrow \mathbb{R}$ be given by $T(y) = f'(x)y$. Rearranging the definition of the derivative,

$$\lim_{\Delta x \rightarrow 0} \frac{|f(x + \Delta x) - f(x) - T(\Delta x)|}{|\Delta x|} = 0$$

- The derivative of f at x is the linear map T satisfying this limit

The derivative as a linear map

- It further follows that $f' : \mathbb{R} \rightarrow \mathbb{R}$ is really a function from $\mathbb{R}$ to the linear maps from $\mathbb{R}$ to $\mathbb{R}$, i.e. $f' : \mathbb{R} \rightarrow \mathbb{R}^*$
 - Of course $\mathbb{R}^* \approx \mathbb{R}$, which is why we usually think of $f' : \mathbb{R} \rightarrow \mathbb{R}$
 - But this will become an issue for functions from $\mathbb{R}^n \rightarrow \mathbb{R}$

Total differentials

- Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and write $y = f(x_1, x_2)$
- What happens to y when x_1 changes by dx_1 and x_2 is held fixed?
 - From our univariate work, in differential form, $dy = f_1 dx_1$
- Now what happens if x_2 also changes, by dx_2 ?
 - The corresponding differential is $dy = f_1 dx_1 + f_2 dx_2$
 - Intuitively, the total change in f is the sum of the changes in f resulting from changes in $\mathbf{x}$ along each coordinate axis

Total differentials

- More generally, if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ then

$$df = \sum_{i=1}^n f_i dx_i = f_i dx_i$$

- Written this way, df is the *total differential* of f
- Notice that while the arguments have been ignored, df is a local notion
 - The total differential of f depends on $\mathbf{x}$ since all of the f_i depend on $\mathbf{x}$
- The total differential is an extremely useful tool for gaining intuition and deriving results concerning optimization and comparative statics

The derivative

- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and let $d\mathbf{x}$ be a column vector of differentials, $d\mathbf{x} = (dx_1, \dots, dx_n)'$
- Let Df be the row vector of partial derivatives (*linear functional*)

$$Df = (f_1, \dots, f_n)$$

- ▶ This is the derivative of f
- ▶ It is a local notion, all of the partials are evaluated at a point in $\mathbb{R}^n$
- This row vector would be written ∇f , the gradient vector
 - ▶ Df is also called the *Jacobian derivative* of f at $\mathbf{x}$
 - ▶ When the same partials are arranged as a column we call it the *gradient*, $\nabla f(\mathbf{x})$

The derivative

- Observe that

$$df = \sum_{i=1}^n f_i dx_i = Df d\mathbf{x}$$

- By writing Df as a row vector, and interpreting it as a linear functional acting on $d\mathbf{x}$, we recover the previous intuition
 - The derivative is a linear map Df taking changes $d\mathbf{x}$ in $\mathbf{x}$ to changes df in f

Spaces of linear maps

- To make this formal we need the vector space of linear maps
- If $\mathbf{V}$ and $\mathbf{W}$ are normed vector spaces, let $\mathcal{L}(\mathbf{V}, \mathbf{W})$ be the collection of all linear maps from $\mathbf{V}$ to $\mathbf{W}$
 - Sums and scalar multiples of linear maps are again linear
 - So $\mathcal{L}(\mathbf{V}, \mathbf{W})$ is a vector space

The derivative

- Let $\mathbf{V} = \mathbb{R}^n$ and let $\|\mathbf{v}\|^2 = \sum v_i^2$, be the norm
- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$. Then f is *differentiable* at $\mathbf{x}$ provided there exists a linear map $Df(\mathbf{x}) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ so that

$$\lim_{\|\Delta \mathbf{x}\| \rightarrow 0} \frac{\|f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) - Df(\mathbf{x})(\Delta \mathbf{x})\|}{\|\Delta \mathbf{x}\|} = 0$$

- ▶ $Df(\mathbf{x})$ is the *derivative* of f at $\mathbf{x}$
- ▶ The higher-dimensional analog of the univariate limit
- If f is differentiable at all $\mathbf{x}$ in some open $U \subset \mathbb{R}^n$ then $Df : U \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$

The derivative

- Just as before, Df provides a *linear approximation* to f
 - Think of $Df(\mathbf{x})(\Delta\mathbf{x})$ as approximating how the image of f at $\mathbf{x}$ changes when $\mathbf{x}$ changes by $\Delta\mathbf{x}$
 - $\Delta\mathbf{x}$ is now a vector of changes
- In differential form we write $df = Df(\mathbf{x}) d\mathbf{x}$

The matrix of first partials

- We use superscripts to index the range: if $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ then $f = (f^1, \dots, f^m)'$, a column vector of functions $f^i : \mathbb{R}^n \rightarrow \mathbb{R}$

Theorem

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable on some open set and that f_j^i is continuous. Then the derivative Df of f is the matrix of first partials, $Df_{ij} = f_j^i$.

The matrix of first partials

- This corresponds precisely to the observation $df = Df \, dx$
 - $f = (f^i)$ where $f^i : \mathbb{R}^n \rightarrow \mathbb{R}$, so $df = (df^i)$: the change in f is the column of changes in the f^i
 - $df^i = Df^i \, dx$, and $Df^i = (f_1^i, \dots, f_n^i)$ is the i th row of Df
 - Thus $df = Df \, dx$ as desired

Chain rule

Chain Rule

If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^k$ are differentiable then $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is differentiable and

$$Dh(\mathbf{v}) = Dg(f(\mathbf{v})) \circ Df(\mathbf{v})$$

- The chain rule continues to hold, but since we are dealing with matrices, order matters